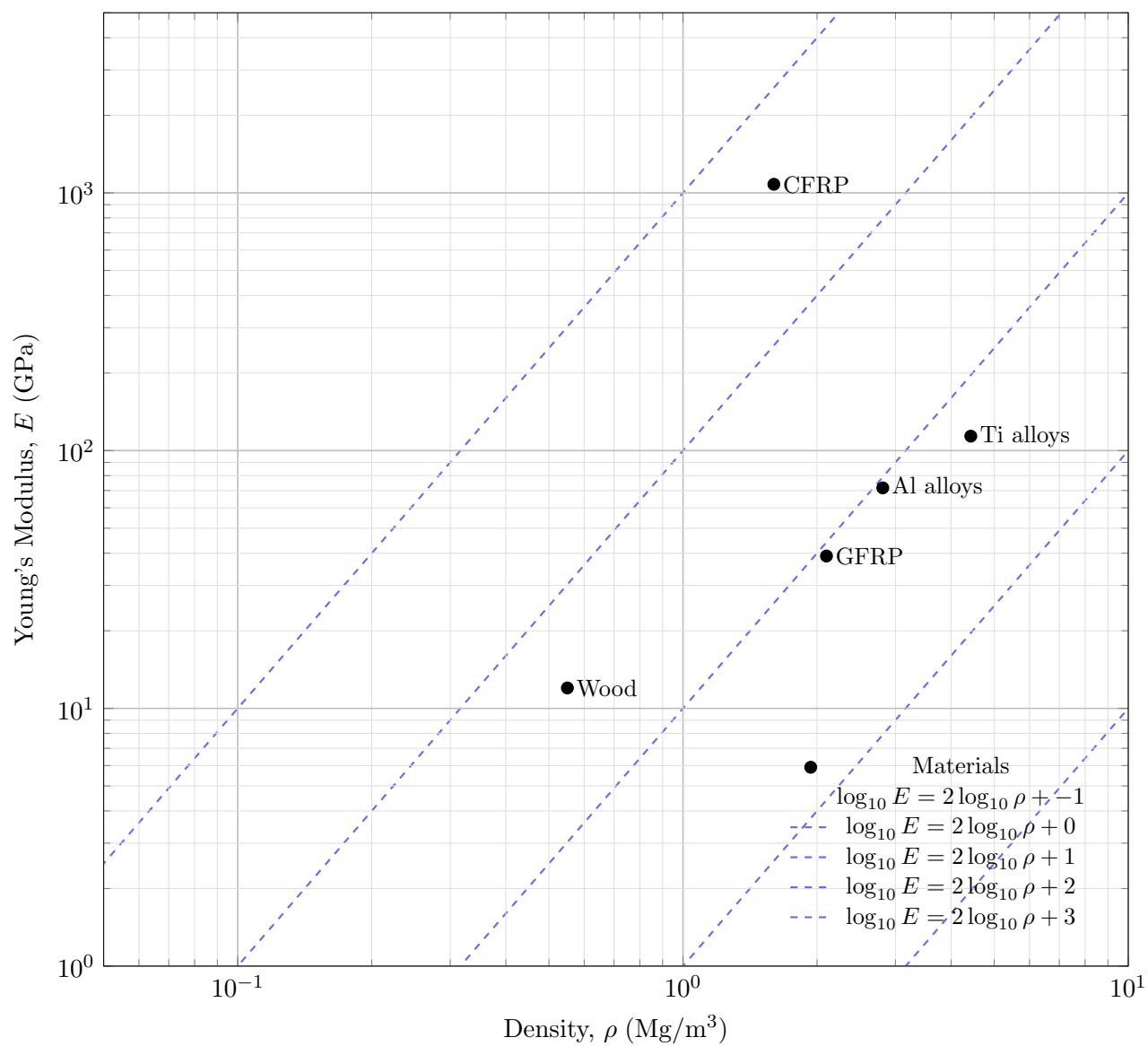




Objective (Maximize):

$$\begin{aligned}\text{Bending stiffness per mass of material} &= \frac{EI}{m}, \quad m = \rho V \\ &= \frac{EI}{\rho V}\end{aligned}$$

$$\text{Material index } M = \frac{E}{\rho}$$

$$\log(M) = \log(E) - \log(\rho)$$

$$\text{Alternative loci: } \log(E) = 2\log(\rho) + \log(\text{const})$$